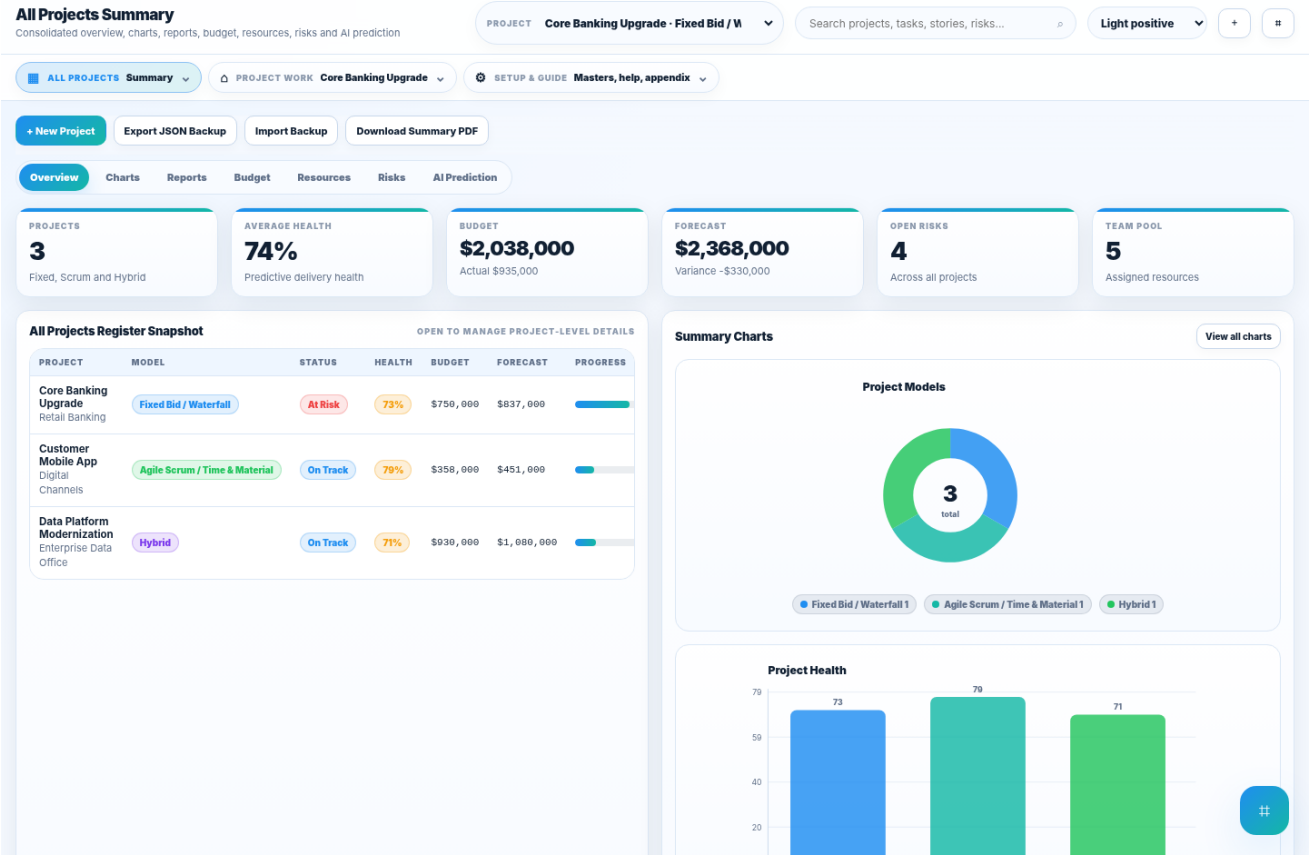


IT Project Manager Workspace

Final single-file offline HTML tool - user guide, report interpretation guide, and codebase whitepaper

This guide explains how to run the single HTML file, manage Fixed Bid/Waterfall, Agile Scrum/T&M; and Hybrid projects, review PMBOK and Agile/Scrum reports, interpret AI Prediction, and understand how the codebase stores data offline.

Interface preview



All Projects Summary uses a top project dropdown, grouped navigation, summary tabs, compact KPIs and hover-enabled custom SVG charts.

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- 6. Agile Scrum / T&M;: Backlog, sprint board and agile metrics
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1. How to Run the HTML File

The application is delivered as one standalone HTML file. It does not require installation, a server, a database, or internet access after download.

- Download the HTML file to your computer.
- Double-click the HTML file, or right-click and open it with a modern browser such as Chrome, Edge, Firefox or Safari.
- The tool loads sample IT project data automatically the first time it opens.
- Your changes are saved inside the same browser profile using local browser storage.
- Use Export JSON Backup before moving to another computer, clearing browser data, or sharing a clean copy.

Browser support

- Recommended: latest stable Chrome, Edge, Firefox or Safari.
- Minimum capability expectation: ES6 JavaScript, SVG, localStorage, CSS variables, CSS Grid and Flexbox.
- Practical baseline: Chrome/Edge 100+, Firefox 100+, Safari 15.4+ or any current enterprise-managed modern browser.
- Internet connection: not required for normal use. No CDN, no external CSS, no external JavaScript and no remote AI service are used.

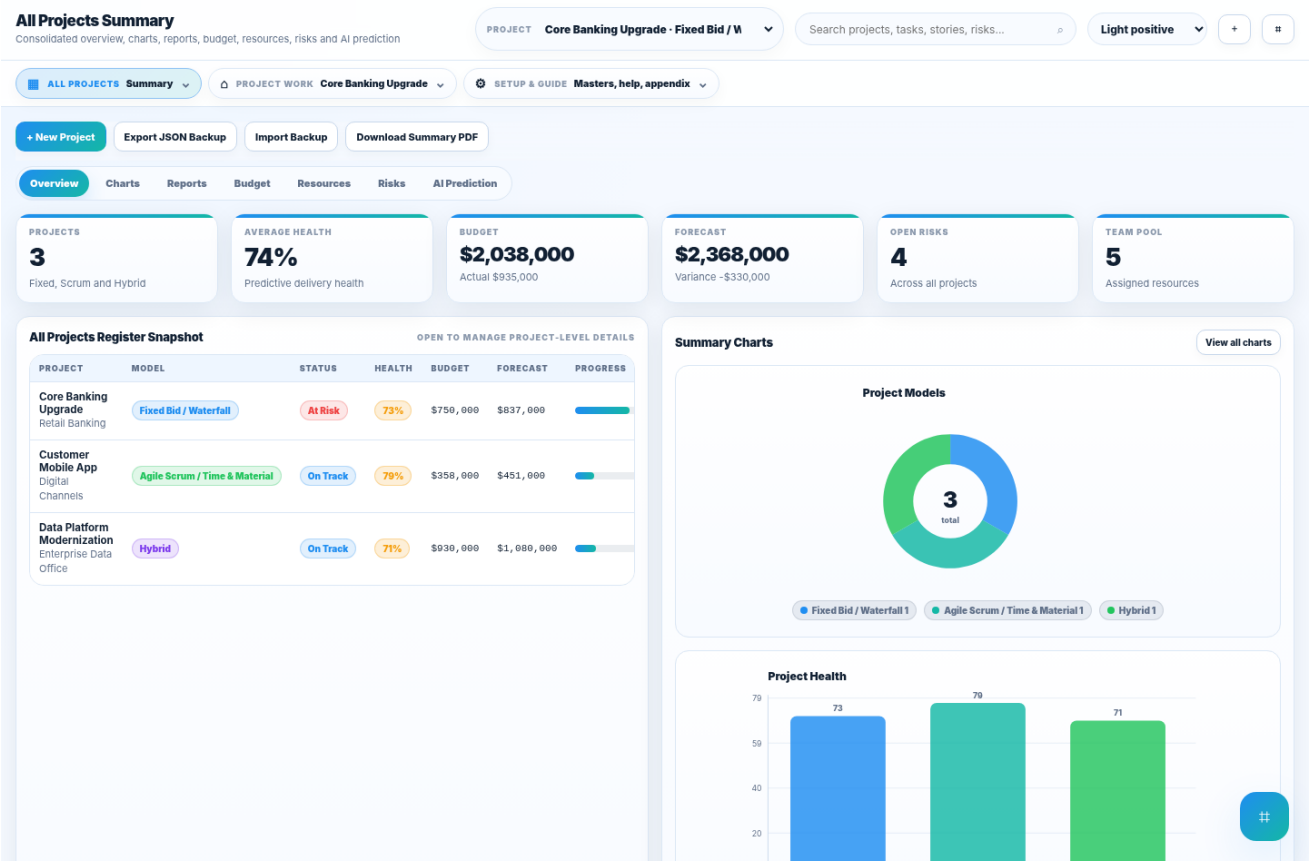
Important local storage note

Data is stored in the current browser profile. If you open the same file in a different browser, private/incognito mode, or another computer, the data will not automatically follow. Use JSON backup and import to transfer data.

2. Interface Structure and Project Selection

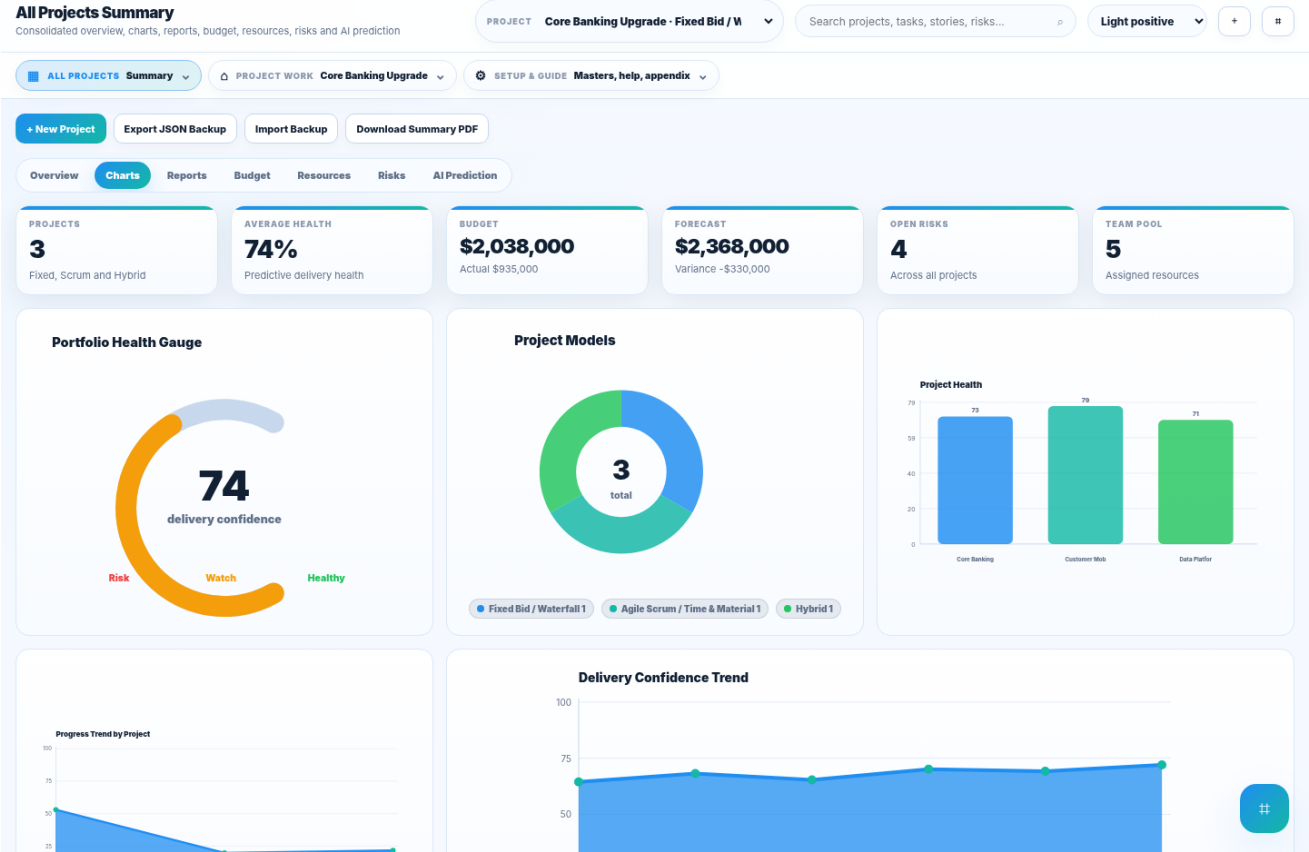
The tool uses a top project dropdown and grouped menus to keep the screen clean. All project-level sections are scoped to the selected project. The All Projects Summary is only for consolidated overview and reporting.

All Projects Summary



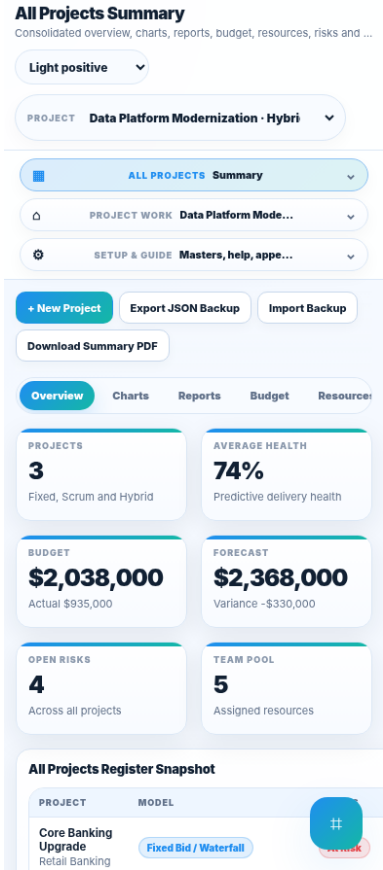
Use Summary only for consolidated KPIs, charts, reports, budget, resources, risks and overall AI Prediction. Do not manage individual project Gantt or sprint items here.

Summary Charts



The summary chart tab mixes gauge, donut, bar, area, grouped, waterfall, bubble, heatmap, radar, funnel, treemap and timeline views.

Mobile responsive layout



On mobile, the interface stacks controls, keeps the project dropdown visible and maintains compact cards for quick review.

3. Choosing the Right Project Operating Model

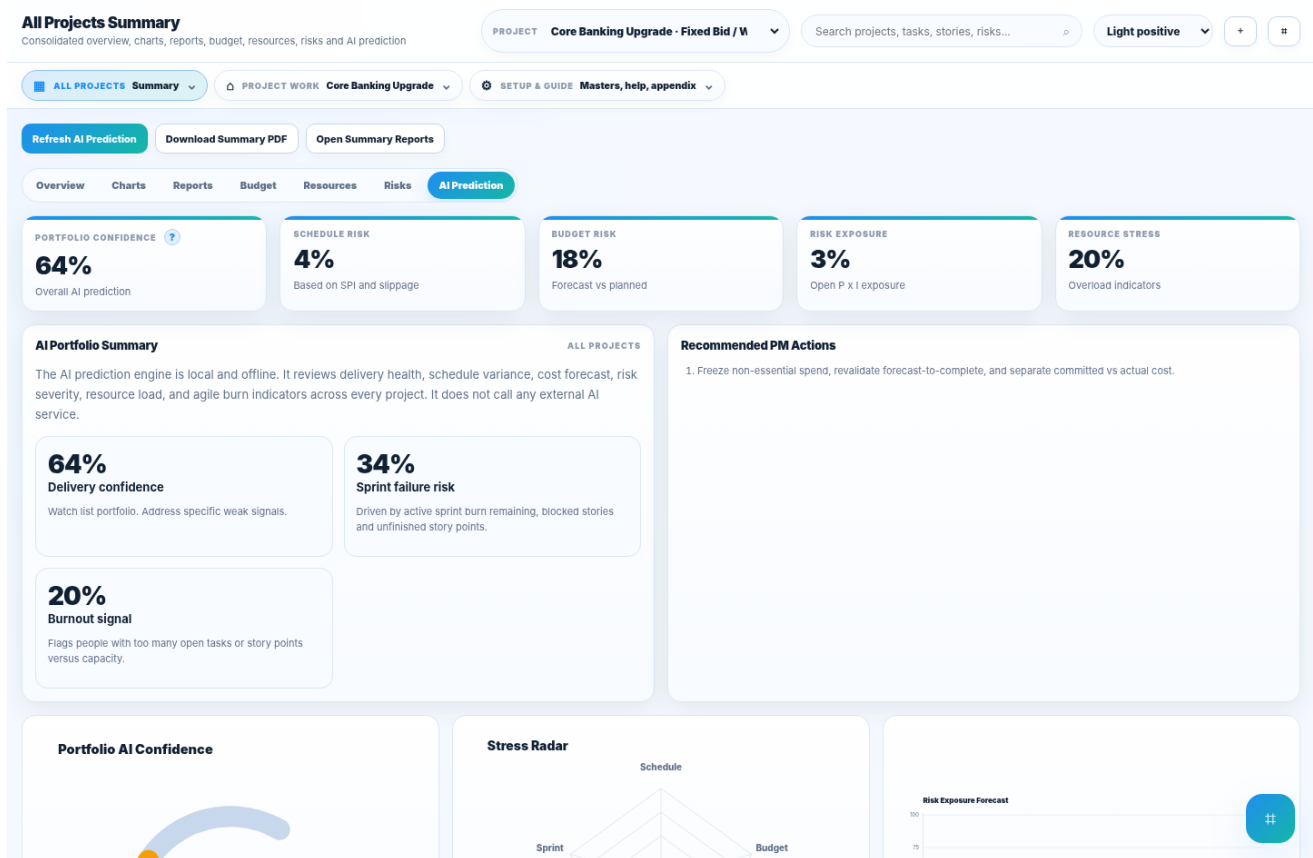
Model	Use when	Primary sections
Fixed Bid / Waterfall	Scope, deliverables, acceptance criteria, cost and schedule are fixed	Fixed Bid / Gantt, Budget, Team, RAID, Reports, AI
Agile Scrum / T&M	Scope evolves sprint-by-sprint and funding is time/material driven	Sprint Board, Backlog, Budget, Team, RAID, Reports, AI
Hybrid	High-level deliverables or milestones are fixed, but execution is agile	Gantt + Sprint Board + Hybrid Links + full reporting

- Changing the selected project from the top dropdown automatically refreshes Project Cockpit, Gantt, Sprint Board, Budget, Team, RAID, Reports and AI Prediction.
- Gantt is disabled for pure Agile projects because the work is managed through Scrum artifacts.
- Sprint Board is disabled for pure Fixed Bid projects because scope is managed through WBS tasks and baselines.
- Hybrid projects show both work management styles and provide Hybrid Links for traceability.

4. All Projects Summary and Overall AI Prediction

- Overview: executive KPIs and project register snapshot.
- Charts: all-project chart wall with multiple chart types.
- Reports: consolidated management report and attention table.
- Budget: planned, actual, forecast and variance across projects.
- Resources: assigned resource pool and utilization signals.
- Risks: cross-project risk register and exposure summary.
- AI Prediction: portfolio confidence, stress radar, risk forecast, budget forecast and recommended PM actions.

Overall AI Prediction



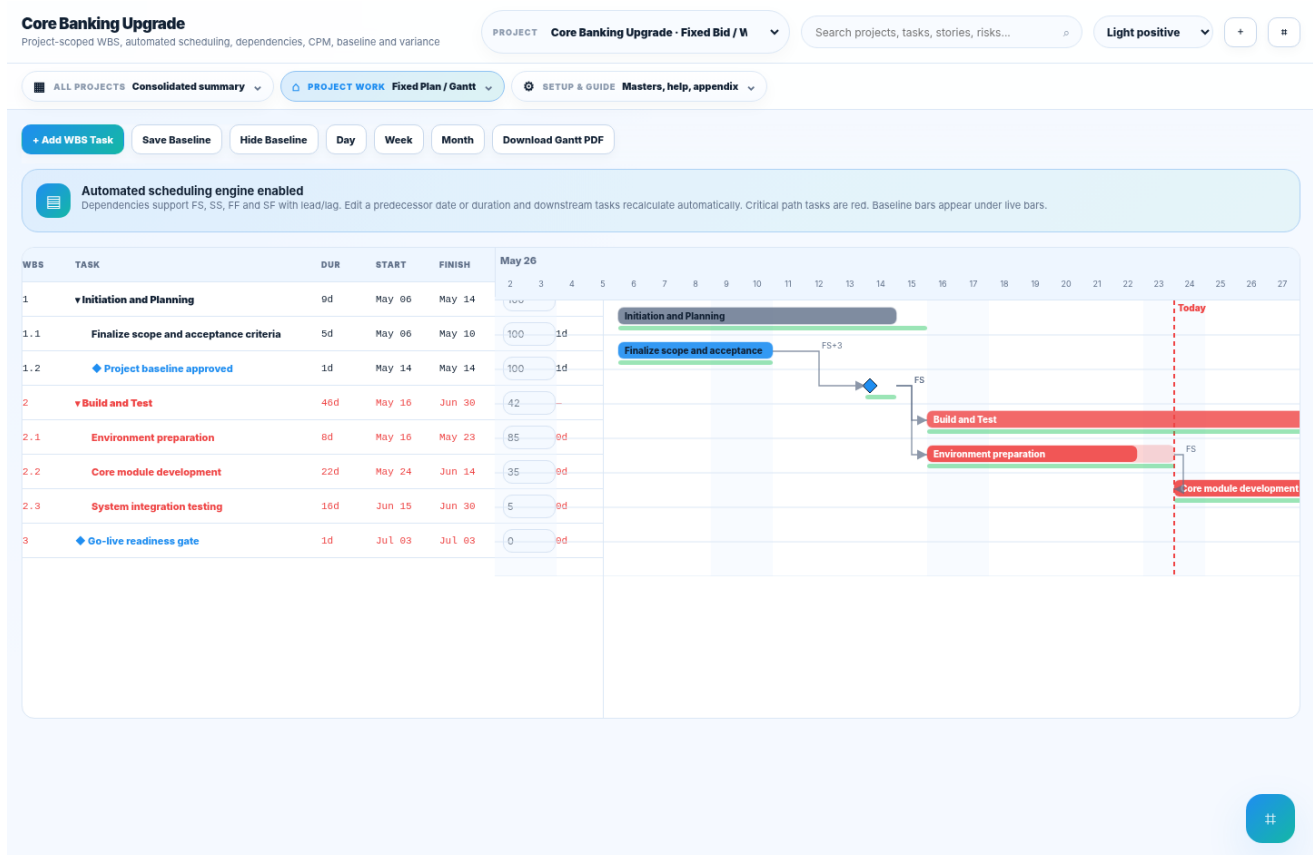
Overall AI Prediction combines schedule risk, budget risk, risk exposure, resource stress and sprint burn risk across all projects.

5. Fixed Bid / Waterfall: WBS, Dynamic Gantt, Dependencies and Baseline

For fixed-bid and hybrid projects, the Gantt section behaves as a project-scoped scheduling workspace. It is not a static image. It is driven by a scheduling engine.

- Create WBS tasks, summary tasks and milestones.
- Use dependency strings such as 1.2FS+2, 2.1SS+0, 3FF-1 or 4SF+3.
- When a predecessor moves, downstream tasks recalculate automatically.
- Critical Path Method logic calculates total float. Zero-float tasks are highlighted in red.
- Save Baseline after approval. Baseline bars remain visible underneath live bars to show slippage.
- Use the Gantt PDF button to print or save the selected-project Gantt.

Dynamic Gantt and WBS



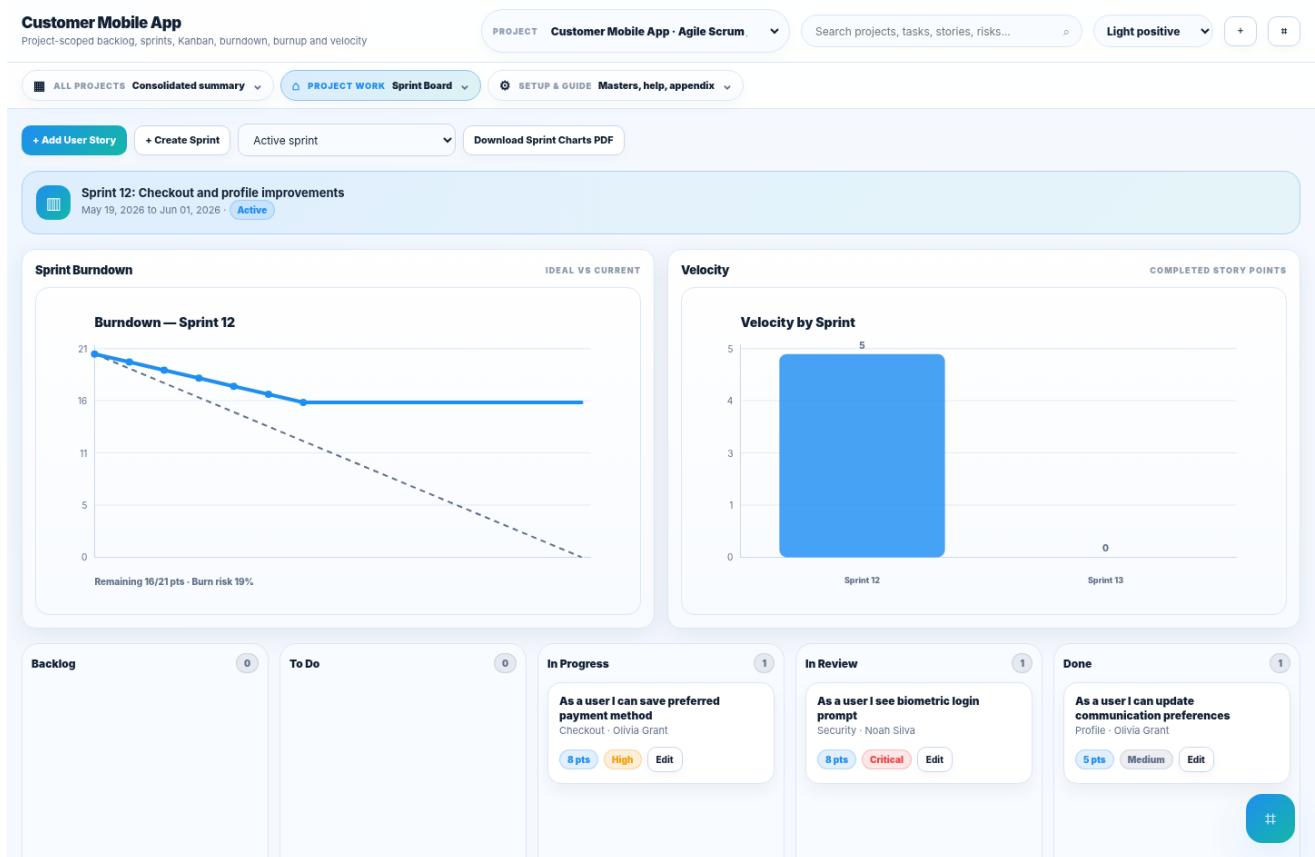
The Gantt view shows WBS on the left, live bars on the right, baseline overlays, dependency lines, critical path and today marker.

6. Agile Scrum / T&M;: Backlog, Sprint Board and Metrics

For Agile Scrum/T&M; and Hybrid projects, the Sprint Board supports sprint planning, backlog grooming and Kanban-style execution.

- Create sprints and define sprint goals.
- Create user stories with epic, story points, priority, assignee and blocker flag.
- Drag stories across Backlog, To Do, In Progress, In Review and Done.
- Review burndown, burnup, velocity and cumulative flow charts in reports.
- Use AI Prediction to detect sprint failure risk based on remaining points and blockers.

Sprint board



The Scrum board is available for Agile Scrum/T&M; and Hybrid projects. It stays scoped to the selected project.

7. Hybrid Delivery and Traceability

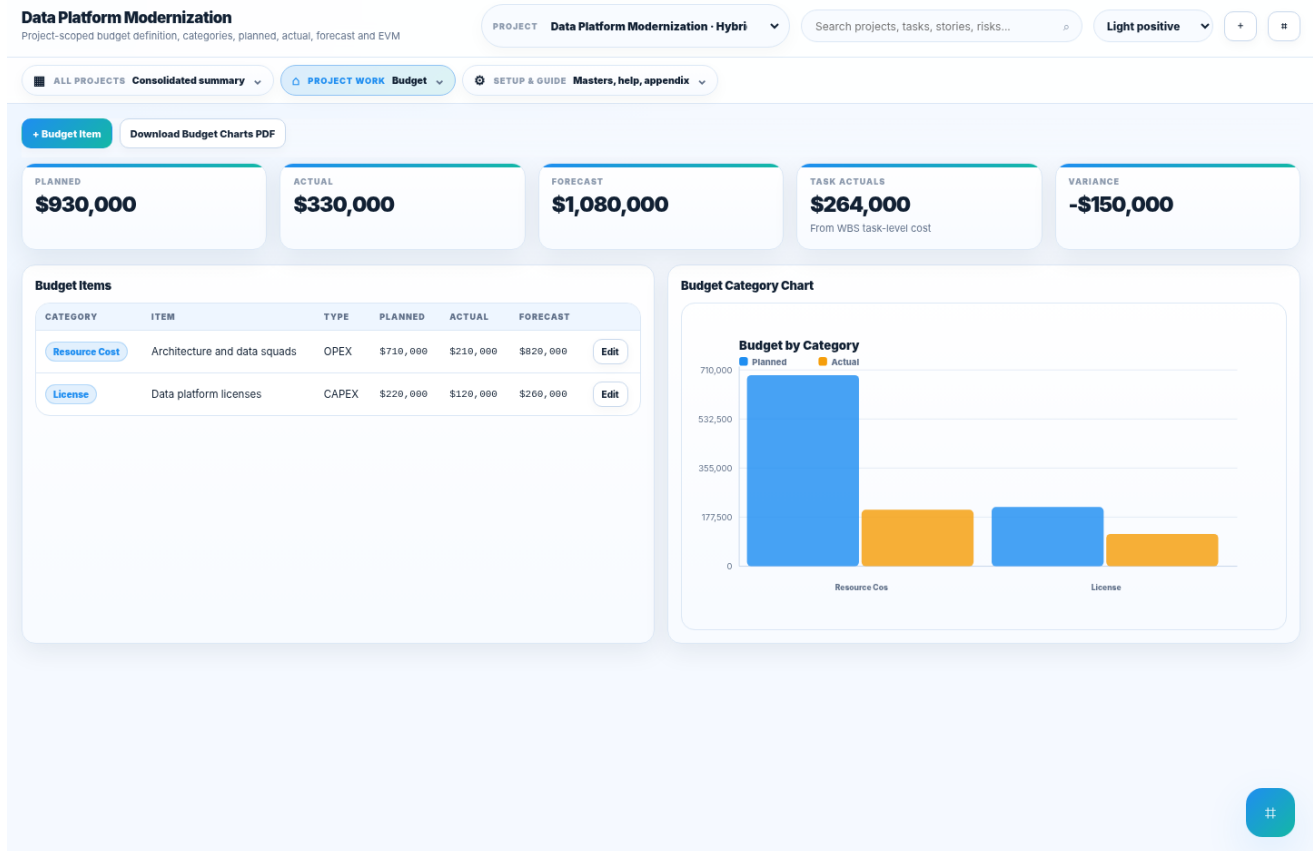
Hybrid projects combine fixed governance deliverables with agile execution stories. The Hybrid Links section connects WBS items to Scrum stories.

- Use WBS/Gantt to manage fixed milestones, deliverables, baselines and governance checkpoints.
- Use Sprint Board to manage the detailed delivery work in stories and sprints.
- Use Hybrid Links to connect a story to the WBS deliverable it supports.
- Review reports to find fixed deliverables that have no linked agile stories or blocked stories that threaten a fixed milestone.

Hybrid best practice: keep governance at deliverable level and agile execution at story level. Do not duplicate the same work as both a full WBS task and a full story unless you intentionally need traceability.

8. Budget, Team and RAID Management

Project budget section



Budget is selected-project only. It includes planned, actual and forecast cost, budget categories, and EVM-style indicators.

Budget workflow

- Add budget line items for resource cost, license, infrastructure/cloud, vendor, training, travel, contingency and other categories.
- Track planned, actual and forecast cost separately.
- Review budget variance and EVM indicators inside Budget and Project Reports.
- Use budget waterfall and planned/actual/forecast charts to explain the financial story.

Team workflow

- Maintain the resource master with name, designation, location, skills, rate and capacity.
- Select resources for each project so project reports show only the selected project team.
- Review team cost treemap, capacity chart and utilization bubble chart.
- Quick-add resources from task/story creation when planning without leaving the flow.

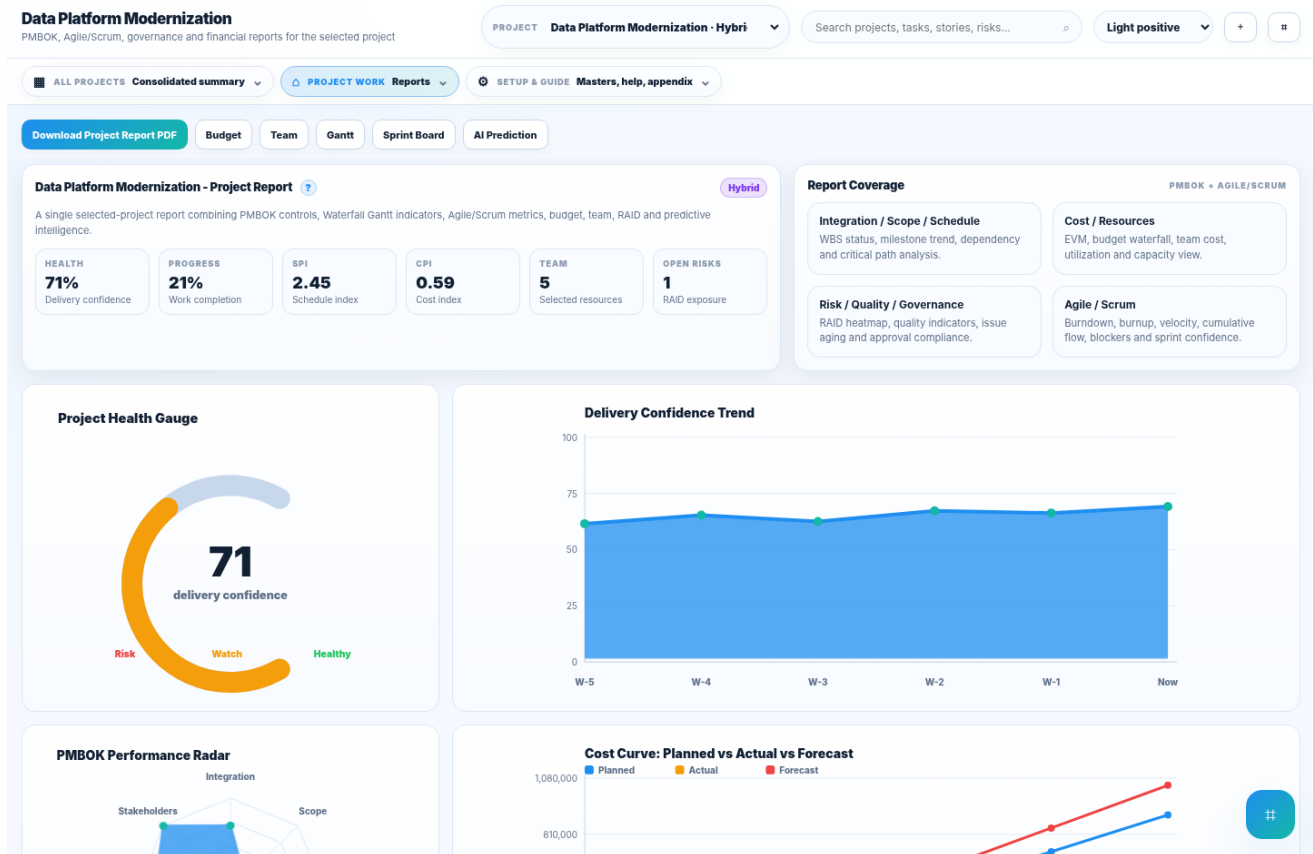
RAID workflow

- Log risks with probability, impact, category, owner and response plan.
- Review 5x5 matrix, bubble chart, issue aging and risk exposure.
- Escalate high probability x impact items and track mitigation owners weekly.

9. Project-Level Reports and Chart Interpretation

Project Reports are scoped only to the active project in the top dropdown. They include PMBOK, Waterfall, Agile/Scrum, Hybrid, budget, team, risk and AI-related insights.

Selected-project report



Project Reports combine summary KPIs, coverage cards, PMBOK radar, cost curves, budget waterfall, work status, team charts, risk charts, agile charts and Gantt-specific indicators.

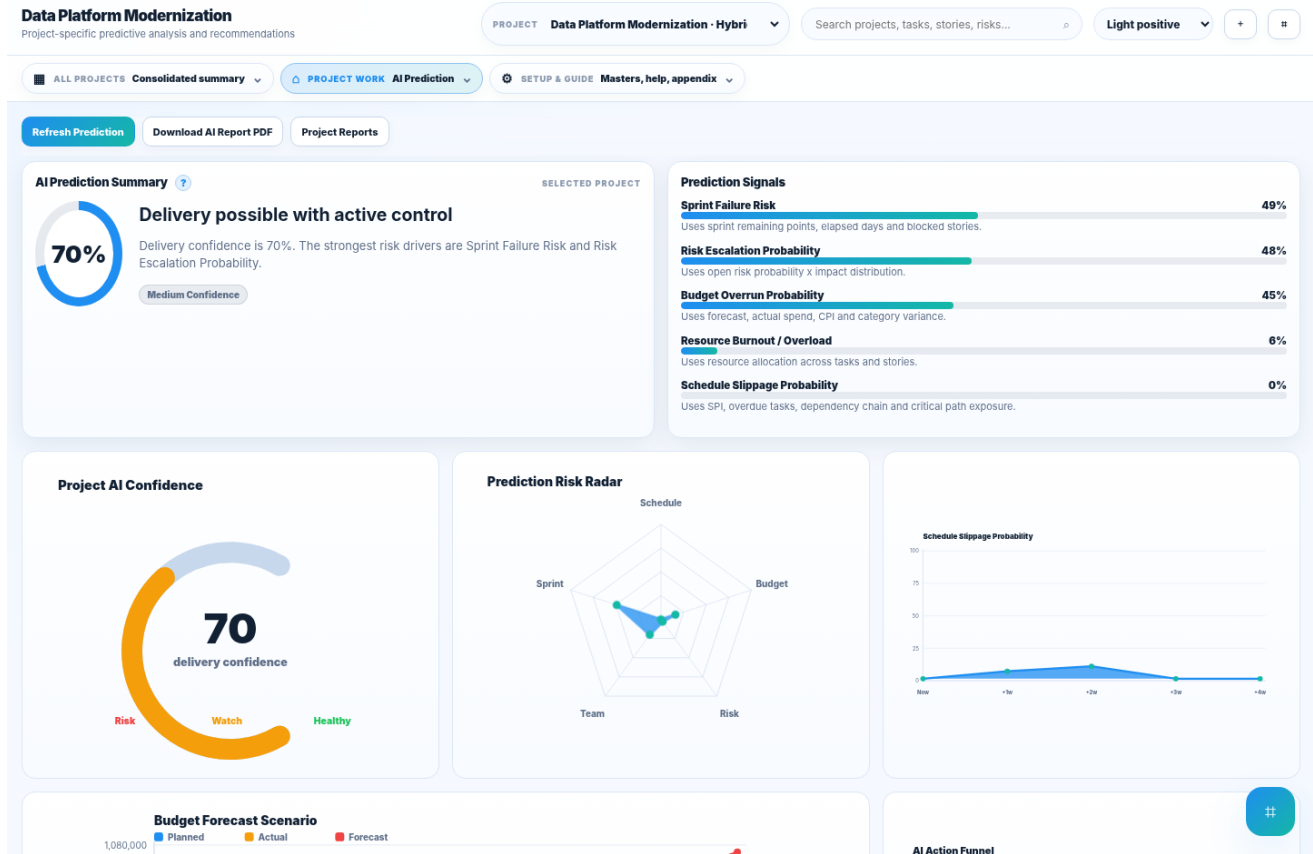
Report categories

- Integration and scope: work item report, WBS status, hybrid traceability and governance checkpoints.
- Schedule: SPI, milestone timeline, critical path, task float and dependency network.
- Cost: CPI, budget line detail, cost curve, budget waterfall and forecast variance.
- Resources: team table, capacity chart, utilization vs cost bubble and cost treemap.
- Risk/quality/governance: risk heatmap, risk bubble, issue aging, quality trend, PMBOK radar and compliance radar.
- Agile/Scrum: burndown, burnup, velocity and cumulative flow for applicable project models.

10. AI Prediction: Project and All-Project Intelligence

AI Prediction is a local, offline, JavaScript-based scoring engine. It does not call an AI service. The purpose is to provide early warning and recommendations based on project data.

Selected-project AI Prediction



Selected-project AI Prediction reviews schedule, budget, risk, team and sprint signals for the active project only.

AI signals used

- Schedule: SPI, overdue tasks, critical path, dependency signals and baseline slippage.
- Budget: planned vs actual vs forecast, CPI and cost variance.
- Risk: open risk count and probability x impact severity.
- Resources: utilization, overload and concentration of work on the same person.
- Agile: remaining story points, blocked stories, burn risk and sprint progress.
- Output: confidence score, prediction radar, trend forecast, action funnel and smart PM recommendations.

11. Masters, Help, Appendix and Codebase Whitepaper

Help and workflow guide

Help and Workflow Guide

Detailed workflow, screenshots, do/don'ts and best practices

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Data Platform Modernization · Hybri

Search projects, tasks, stories, risks...

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ALL PROJECTS

Consolidated summary

PROJECT WORK

Data Platform Modern...

SETUP & GUIDE

Help

Start-to-Finish Workflow

HOW TO USE

1

Create the project.

Choose Fixed Bid / Waterfall, Agile Scrum / T&M, or Hybrid. This controls whether Gantt, Scrum board, or both are available.

2

Select the project from the top dropdown.

All project-level sections refresh from this selected project only: Gantt, Sprint Board, Budget, Team, RAID, Reports and AI Prediction.

3

Plan work by model.

Fixed projects use WBS tasks, dependencies, CPM and baseline. Agile projects use backlog, sprint planning and Kanban. Hybrid projects use both with traceability links.

4

Define budget and team.

Add project-level budget lines and assign resources from the master list. Resources can also be created quickly while planning work.

5

Review PMBOK, Agile and AI reports.

Use Project Reports for selected-project reporting. Use Summary tabs for all-project control-tower reporting.

Interface Map

SCREENSHOT GUIDE

All Projects Summary

Use only for consolidated health, charts, budget, resources, risks and overall AI prediction.

Selected Project Work Area

Use the top dropdown first, then open Gantt, Sprint Board, Budget, Team, RAID, Reports or AI.

Reports and Charts

Hover charts to view exact values. Use PDF buttons to print or save reports.

Do's and Don'ts

Do

- Pick the correct delivery model before planning.
- Keep WBS dates, dependencies and baseline current.
- Review SPI, CPI, risk heatmap and AI prediction weekly.
- Separate planned, actual and forecast budget values.
- Use Hybrid Links when sprint stories deliver

Don't

- Do not track selected-project work inside summary views.
- Do not use Gantt for Scrum-only work unless the project is Hybrid.
- Do not ignore blocked stories, overdue tasks or zero-float critical path tasks.
- Do not overwrite baseline after slippage without governance approval.
- Do not treat AI prediction as final truth; use it

Industry Best Practices

- PMBOK alignment: maintain charter, scope baseline, schedule baseline, cost baseline, risk register, change control and stakeholder communications.
- Waterfall: confirm acceptance criteria, dependencies, critical path, baseline and formal approvals.
- Scrum: keep the product backlog ordered, plan sprint goals, maintain WIP discipline and review burndown daily.
- Hybrid: do governance at deliverable level and execution at story level; keep traceability explicit.
- Reporting: use a mix of trend, matrix, radar, funnel, heatmap, gauge, waterfall and table views. One chart type never tells the full story.

The Help section provides step-by-step workflow guidance, interface map, do and do not guidance, and best practices.

Calculation appendix

Appendix

Calculation formulas, scoring logic and interpretation guide

PROJECTData Platform Modernization - Hybri

Search projects, tasks, stories, risks...

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PROJECT WORKData Platform Modern...

SETUP & GUIDEAppendix

Formula and Calculation Appendix

PMBOK + AGILE/SCRUM + AI SCORING

Earned Value (EV)

EV = Approved Budget x Percent Complete

Represents the value of completed work.

Planned Value (PV)

PV = Approved Budget x Planned Elapsed Percent

Represents the budgeted value of work planned by today.

Actual Cost (AC)

AC = Sum of actual budget line items

Actual cost spent or recorded so far.

SPI

SPI = EV / PV

Below 1.00 indicates schedule underperformance.

CPI

CPI = EV / AC

Below 1.00 indicates cost inefficiency.

Schedule Variance

SV = EV - PV

Negative value means behind planned progress.

Cost Variance

CV = EV - AC

Negative value means cost overrun.

Estimate at Completion

EAC = Forecast cost, or BAC / CPI when forecast is not defined

Used for budget overrun prediction.

Risk Score

Risk Score = Probability x Impact

Usually 1-25 on a 5 x 5 matrix.

Critical Path Float

Total Float = Latest Start - Earliest Start

Zero-float tasks are critical and highlighted in the Gantt.

Dependency Logic

FS, SS, FF, SF with lead/lag, e.g. 1.2FS+2

The scheduling engine recalculates successor dates when predecessors move.

Baseline Slippage

Slippage = Current Finish - Baseline Finish

Baseline bars are displayed under live Gantt bars.

Velocity

Velocity = Sum of completed story points in sprint

Used for sprint and release forecasting.

Burndown

Remaining Points = Total Sprint Points - Done Points

Actual remaining points are compared against the ideal burn line.

Burnup

Completed Points over Time vs Scope Line

Shows delivery growth and scope movement.

AI Confidence

Confidence = 100 - schedule penalty - cost penalty - risk penalty - overload penalty - sprint penalty

A local rule-based score. It is not an external AI call.

Report Interpretation Guide

Gauge

Quick executive health. Use it for a fast status read, not root-cause analysis.

Radar

Compares multiple PMBOK or agile dimensions at once.

Waterfall

Explains how budget moves from planned to actual, forecast and variance.

Heatmap

Highlights concentration of risk or governance stress.

Bubble

Shows three dimensions such as probab impact and severity together.

The Appendix explains formulas for EVM, critical path, dependencies, baseline slippage, agile metrics and AI confidence scoring.

Codebase whitepaper

Codebase Whitepaper

Technology, storage, architecture, export and offline behavior

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ALL PROJECTSConsolidated summary

PROJECT WORKData Platform Modern...

SETUP & GUIDECode Whitepaper

Codebase Whitepaper

SINGLE-FILE OFFLINE APPLICATION

Technology Stack

HTML5, CSS3 and vanilla JavaScript. No external CDN, no external chart library, no server, and no internet call. Charts are rendered using custom inline SVG and DOM logic, which keeps the tool portable and fully offline.

Single HTML Structure

- **CSS:** theme variables, responsive layout, dashboard cards, menus, charts, modals and print styles.
- **Data Layer:** local browser persistence using localStorage, with JSON export/import backup.
- **Domain Model:** projects, resources, WBS tasks, sprints, stories, budgets, RAID risks and master data.
- **Scheduling Engine:** dependency parser, date shifting, CPM float calculation, critical path flagging and baseline overlay.
- **Reporting Engine:** custom SVG charts, tables, filters, hover tooltips and browser print/PDF export.
- **AI Engine:** local rule-based predictive scoring for scheaule, budget, risk, resource and sprint confidence.

Data Storage

Data is saved in the browser under a localStorage key. The file can be opened offline in a modern browser. Data remains available after refresh/reopen in the same browser profile. JSON backup should be exported regularly before sharing or moving devices.

Third-Party Tools

None in the final single-file tool. Earlier prototypes may have used charting ideas, but this version uses custom SVG charts so the tool can be shared publicly without CDN or package dependency.

Browser Support

Recommended: latest Chrome, Edge, Firefox or Safari. Required features: ES6 JavaScript, localStorage, SVG, CSS Grid/Flexbox and modern CSS variables. The application does not require internet connectivity after download.

Security and Privacy

All project data stays on the user's computer/browser. The tool does not transmit data. Users should still avoid storing confidential client data in shared browsers and should use encrypted storage/device policies where needed.

The Codebase Whitepaper explains the technology stack, data storage, chart rendering, offline behavior, privacy and browser support.

12. Backup, Restore, PDF Export and Troubleshooting

Backup and restore

- Export JSON Backup regularly. This is the safest way to preserve or transfer project data.
- Import Backup restores previously exported JSON data.
- Before sharing the HTML publicly, reset sample data or share a clean copy without confidential data.
- Browser storage is profile-specific. Clearing site data will remove saved projects unless you have a JSON backup.

PDF export

- Use the Summary PDF button for all-project summary reporting.
- Use Project Report PDF from the selected-project report page.
- Use Gantt PDF from the selected-project Gantt page.
- The tool uses browser print/save as PDF. Choose landscape mode for wide Gantt or chart pages if needed.

Troubleshooting

- If data does not appear, confirm you are opening the same browser profile where the data was saved.
- If popups are blocked during PDF export, allow popups for the local file and retry.
- If charts look narrow, increase browser zoom out slightly or widen the window. The tool also supports mobile stacked layout.
- If the Gantt does not move as expected, inspect predecessor dependency syntax and ensure task dates are valid.
- If Scrum charts are empty, create at least one sprint and add stories with story points.

13. Do's, Don'ts and Best Practices

Do

- Choose the correct project operating model at project creation.
- Keep project-level work inside the selected-project sections, not the summary section.
- Review Summary AI weekly for cross-project stress.
- Review Project AI before status meetings and steering meetings.
- Use baseline only after formal schedule approval.
- Keep risk mitigation owners and target dates current.
- Use Hybrid Links to preserve traceability between agile stories and fixed deliverables.

Do not

- Do not use the All Projects Summary for WBS, sprint or budget line entry.
- Do not overwrite baseline just to hide slippage.
- Do not ignore critical path tasks with zero float.
- Do not use one chart type to make delivery decisions. Combine trends, heatmaps, tables and variance views.
- Do not treat AI Prediction as a replacement for PM judgment. Use it as an early-warning assistant.

Industry-aligned best practices

- PMBOK: maintain scope baseline, schedule baseline, cost baseline, RAID, change control and stakeholder communication cadence.
- Waterfall: use WBS decomposition, dependency control, baseline variance, critical path and formal acceptance checkpoints.
- Agile/Scrum: maintain backlog order, sprint goals, WIP discipline, daily blocker management, burndown and retrospective action tracking.
- Hybrid: govern at milestone/deliverable level while executing detailed work in sprints with traceability.
- Reporting: use KPIs for quick read, charts for patterns, tables for auditability and AI Prediction for early warning.

Formula Reference

Metric	Formula	Interpretation
EV	$EV = BAC \times \text{percent complete}$	Value of work actually earned.
PV	$PV = BAC \times \text{planned elapsed percent}$	Value of work planned by today.
AC	AC = sum of actual cost	Actual spend to date.
SPI	$SPI = EV / PV$	Below 1 means behind schedule.
CPI	$CPI = EV / AC$	Below 1 means over budget or cost inefficient.
SV	$SV = EV - PV$	Negative means schedule variance.
CV	$CV = EV - AC$	Negative means cost variance.
Risk score	Probability x Impact	Higher score needs mitigation/escalation.
Float	Latest Start - Earliest Start	Zero float means critical path.
Slippage	Current Finish - Baseline Finish	Positive days show schedule slip.
Velocity	Completed story points in sprint	Used for sprint/release forecasting.
Burndown	Total points - done points	Remaining work vs ideal sprint line.
AI confidence	100 - risk penalties	Local early-warning confidence score.

End of guide.